
Quiet Orbit QO-R1: Summary

copperhead

<https://github.com/copperheadhq/copperhead-quiet-orbit>

A USB-C powered four-LED ambient lamp board, 57×57 mm

Abstract

QO-R1 is a two-layer, 57×57 mm board that drives four amber LEDs from an ATtiny84A, each on its own hardware PWM channel, powered from USB-C at 5 V. This note lists the hard facts: key figures, channel map, bill of materials, board statistics and the checks required before ordering. The board has not been fabricated, the firmware has not been compiled, and every part number is unverified. The full description is in the companion report, `QO-R1-reference.pdf`.

Status: not built, not tested.

- No board has been fabricated or powered.
 - The firmware has not been compiled.
 - Every part number in the BOM is unverified.
 - D1 (1N4148W) is the wrong part. Fit a 1 A Schottky (B5819W or 1N5819HW class) in the same SOD-123 footprint.
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1 Key figures

Parameter	Value
Board	57×57 mm, 2 layers, 1.6 mm, all parts on top
Input	USB-C, 5 V sink only; 5.1 kΩ on CC1 and CC2; no data, no PD
Protection	100 mA PTC (F1), series diode (D1)
MCU	ATtiny84A, SOIC-14, internal 8 MHz RC oscillator
LEDs	4 × amber, 1206, 590 to 605 nm, active-low
LED current	2.26 to 4.38 mA each (calculated, 680 Ω)
PWM	4 hardware channels, 8-bit, ≈488 Hz
Fade	Triangle, ≈1.0 s period, channels 90° apart
Supply current	30.6 mA calculated maximum; requirement <100 mA
LED positions	Corners of a 40 mm square centred on the board
Mounting	4 × M3, 3.2 mm NPTH, centres 4 mm from each edge
Programming	2×3 AVR ISP, 2.54 mm; powered from USB, ISP VCC is sense only
Enclosure	80×80×25 mm, four light windows

Table 1: Key figures.

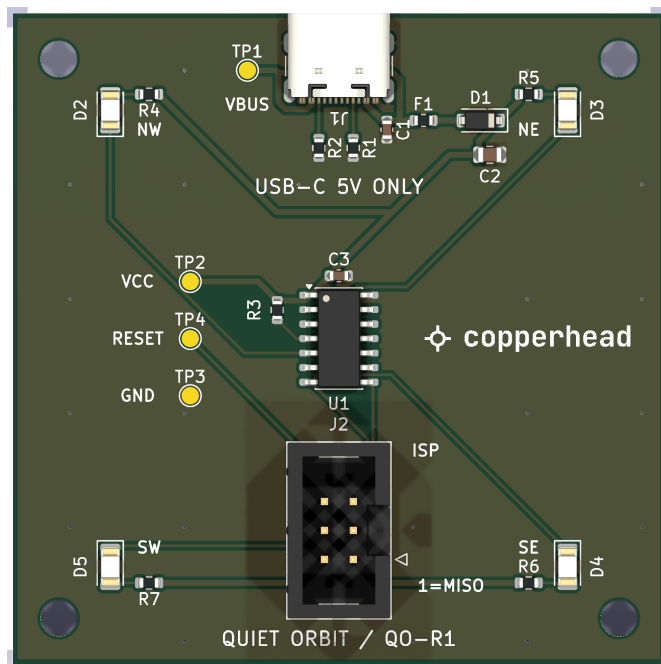


Figure 1: Top side. KiCad render, not a photograph.

2 LED channels

Position	LED	Resistor	U1 pin	PWM output	Shared with
NW	D2	R4	5 (PB2)	OC0A	none
NE	D3	R5	6 (PA7)	OC0B	none
SE	D4	R6	7 (PA6)	OC1A	ISP MOSI
SW	D5	R7	8 (PA5)	OC1B	ISP MISO

Table 2: LED channel map. SCK (pin 9) and RESET (pin 4) carry no LED; RESET has a 10 k Ω pull-up.

3 Schematic

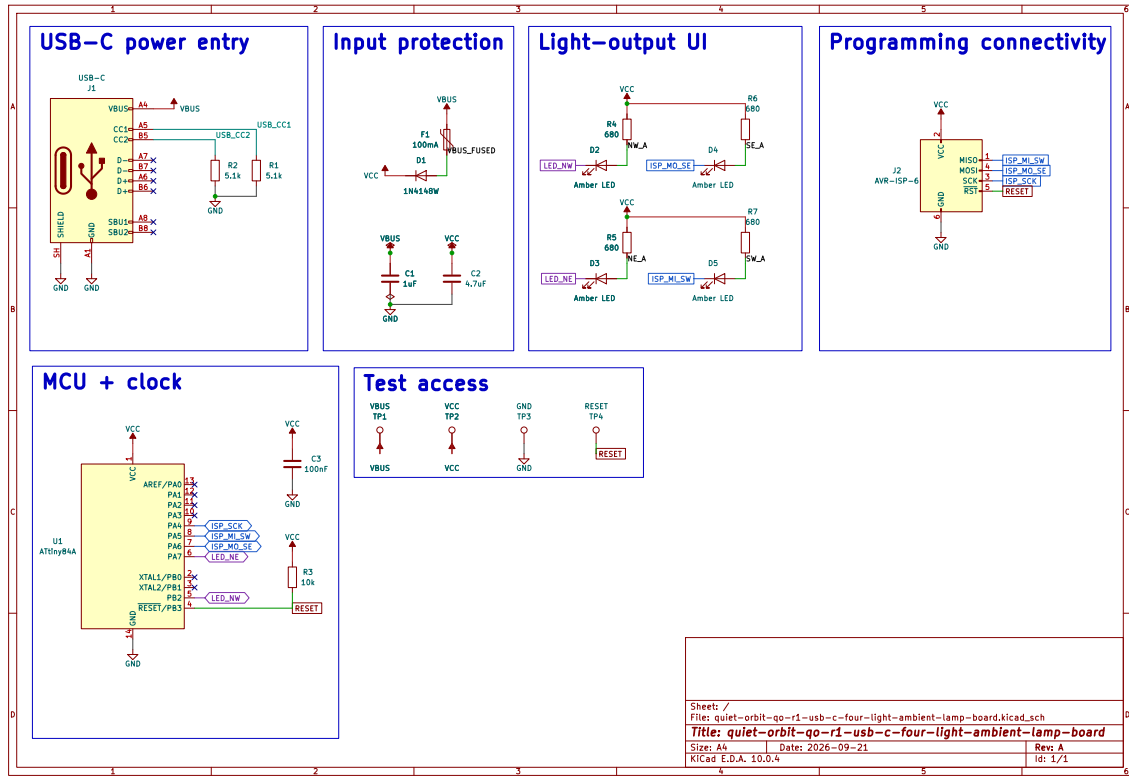


Figure 2: Full schematic sheet, scaled to page width. Full size in docs/schematic.pdf.

4 Board as committed

Item	Value
Footprints	27
Tracks	89 segments, 376.5 mm (320.8 mm top, 55.8 mm bottom)
Vias	24, 0.65/0.3 mm; 21 are GND stitching
Pours	GND, both layers
Rules	0.2 mm clearance; 0.25 mm signal and 0.5 mm power track; 0.4 mm copper to edge
ERC / DRC	0 / 0 violations, 0 unconnected, 0 parity issues (KiCad 10.0.4, all severities)

Table 3: Board statistics, measured from the committed .kicad_pcb.

5 Bill of materials

Ref	Value	Package	MPN (unverified)	LCSC
J1	USB-C receptacle	GCT USB4105, 16P	USB4105-GF-A	C3020560
F1	PTC, 100 mA hold	0603	MF-FSMF010X-2	C210355
D1	1N4148W (replace with 1 A Schottky)	SOD-123	1N4148W-7-F	C81598
C1	1 μ F X7R	0603	CC0603KRX7R9BB105	none found
C2	4.7 μ F X7R	0805	GRM21BR71A475KA73L	none found
C3	100 nF X7R	0603	CC0603KRX7R9BB104	C14663
R1, R2	5.1 k Ω 1%	0603	RC0603FR-075K1L	C105580
R3	10 k Ω 1%	0603	RC0603FR-0710KL	C98220
R4 to R7	680 Ω 1%	0603	RC0603FR-07680RL	C112310
D2 to D5	Amber LED	1206	150120AS75000	C5478408
U1	ATtiny84A	SOIC-14	ATTINY84A-SSUR	C220597
J2	AVR ISP 2 $\times$ 3, shrouded	2.54 mm THT	61200621621	none found
TP1 to TP4	VBUS, VCC, GND, RESET	1.5 mm pad	bare pad	n/a

Table 4: Bill of materials. LCSC codes were matched from a third-party catalogue mirror on 22 September 2026.

6 Checks before ordering

- Replace D1 with a Schottky, as above. The layout does not change.
- Verify every BOM line against a datasheet and a distributor. Three lines have no LCSC match.
- Open the gerbers in a viewer. Nobody has done this yet, and no fab has run a DFM check.
- The J1 footprint puts GND pads 0.194 mm from its NPTH locating pegs. A fab with a 0.2 mm limit will query it.
- CPL rotations are KiCad's and have not been corrected for any assembler.
- Test points are DNP in `docs/BOM.md` but in the BOM in the schematic. One of the two is wrong.
- The LED and M3 hole courtyards are 0.47 mm apart at each corner. An enclosure baffle must fit round the screw head.